Digital Image Processing Technical Report

Author / Team: DIP-Vision.AI Project Team | **Date:** July 04, 2026

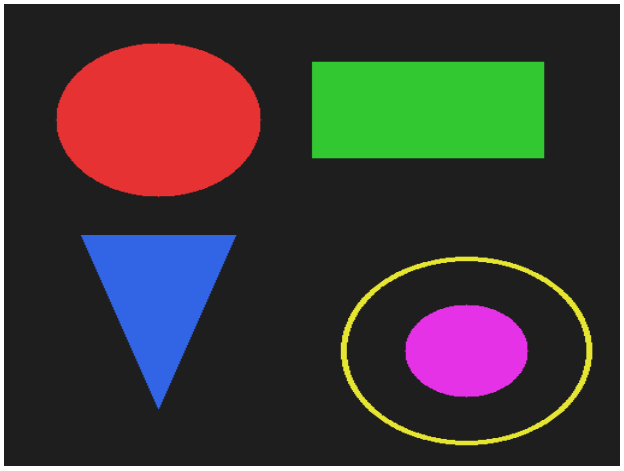
Course: DIP-Vision.AI | Digital Image Processing Toolkit

1. Executive Summary

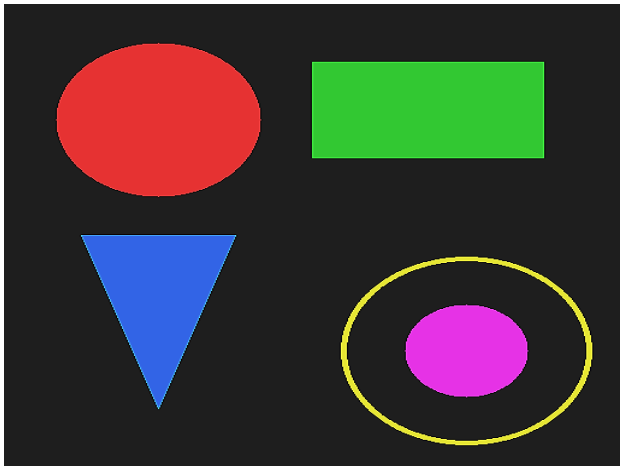
This technical report documents the algorithmic implementation, visual outcomes, and quantitative metrics produced by the modular **Digital Image Processing Toolkit (DIP Toolkit)**. The toolkit encompasses 15 core processing domains including spatial transforms, histogram processing, frequency domain 2D FFT filtering, noise simulation & deblurring restoration, color space manipulation, morphological operations, segmentation, feature extraction, compression, and objective quality evaluation.

2. Visual Comparison & Processing Results

Original Image



Processed Image



3. Quantitative Image Quality Assessment (IQA)

Metric	Value	Interpretation
MSE	23.4368	Mean Squared Error (Lower is closer to original)
RMSE	4.8412	Root Mean Squared Error (L2 Norm intensity deviation)
PSNR (dB)	34.43	Peak Signal-to-Noise Ratio (Higher indicates superior fidelity)
SSIM	0.9762	Structural Similarity Index (1.0 represents perfect structural match)

4. Applied Processing Pipeline

Step #	Operation Domain	Parameters
1	Grayscale & Representation	{'color_mode': 'RGB'}
2	Laplacian Sharpening Filter	{'alpha': 1.0, 'kernel': '3x3'}

5. Mathematical Foundations & Formulations

2D Fast Fourier Transform (FFT):

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j 2\pi (ux/M + vy/N)}$$

Wiener Deconvolution Filter:

$$F\blacksquare(u, v) = [|H(u, v)|^2 / (|H(u, v)|^2 + K)] \cdot [G(u, v) / H(u, v)]$$

Structural Similarity Index (SSIM):

$$SSIM(x, y) = [(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)] / [(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)]$$

6. Additional Engineering Notes

Report compiled programmatically via DIP Toolkit PDF Export Engine.